

Research



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Variations in non-local interaction range lead to emergent chase-and-run in heterogeneous populations

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In a chase-and-run dynamic, the interaction between two individuals is such that one moves towards the other (the chaser), while the other moves away (the runner). Examples can be found in both interacting cells and animals. Here, we investigate the behaviours that can emerge at a population level, for a heterogeneous group that contains subpopulations of chasers and runners. We show that a wide variety of patterns can form, from stationary patterns to oscillatory and population-level chase-and-run, where the latter describes a synchronized collective movement of the two populations. We investigate the conditions under which different behaviours arise, specifically focusing on the interaction ranges: the distances over which cells or organisms can sense one another’s presence. We find that when the interaction range of the chaser is sufficiently larger than that of the runner—or when the interaction range of the chase is sufficiently larger than that of the run—population-level chase-and-run emerges in a robust manner. We discuss the results in the context of phenomena observed in cellular and ecological systems, with particular attention to the dynamics observed experimentally within populations of neural crest and placode cells.

1. Introduction

In cellular and animal systems, interactions frequently trigger movement responses. The collective movements that emerge at a population level have become the focus of considerable interest, in phenomena that range from the swarming and flocking of animals [1], to embryonic development [2] and cancer invasion [3]. Mathematical modelling has helped uncover the mechanistic basis of these emerging dynamics, using methods that range from agent-based (interacting particle systems) to continuous partial differential equation (PDE) systems [4–6].

Attraction and repulsion form two fundamental interaction types, whereby the nearby presence of another individual induces movement towards (attraction) or away from it (repulsion). Within a single homogeneous population, attracting interactions can drive a population to self-organize into an aggregated group, such as a herd of animals, while repelling interactions can enhance the dispersal of an aggregated population. For multiple or heterogeneous populations—a mixture of distinct animal species or cell types, or subpopulations with different traits—greater complexity is possible [7]. In a binary system with two distinct populations, there are four principal

interactions: a set of two homotypic interactions between individuals of the same type and a set of two heterotypic interactions between individuals of different types. A broad set of interaction combinations can be conjured and the question as to how these subsequently translate into population-level dynamics are of manifest interest when it comes to understanding how heterogeneous populations become spatially structured.

Beyond the fundamental nature of an interaction—whether it is attracting or repelling—a second important point of consideration is its range: the distance of separation over which an interaction can occur. These distances will naturally depend on the mechanism through which an interaction is mediated. Animals and cells can use a multitude of mechanisms to sense other individuals [8], through both direct and indirect means. By direct we mean a sense that could (almost) exactly signal the current position of another individual, for example, through directly touching or sighting the neighbour; indirect refers to detection through an intermediary, such as chemical cues or tracks left by the neighbour, that may indicate its recent presence. Whether a sensing is direct or indirect, within heterogeneous groups the interaction ranges of the homotypic and heterotypic interactions could vary considerably: animals range widely with respect to the form and range of their sensory systems—certain species of baleen whales are believed to be able to communicating over 100s of kilometres [9]; different cells can extend a spectrum of cell protrusions from shorter to long range [10].

We will focus here on a particular type of heterotypic interaction, which we call chase-and-run (or pursue-and-escape). Here, individuals of (say) the first population move towards individuals of the second population, which in turn move away. While for the most part, we eschew a specific application in favour of broader insights, we note that such dynamics feature within numerous contexts. In an ecological context, the obvious example would be between a predator and its prey [11], but it could also occur between dominant and submissive members of a pack. At the cellular level, chase-and-run has been observed both *in vitro* and *in vivo* for a variety of heterogeneous cell groups, such as between distinct zebrafish pigment cell types [12] or between embryonic neural crest and placode populations [13].

The case of neural crest cells (NCC) and placode cells (PC) interactions offers a particularly illuminating exemplar in the context of the objectives here. Chase-and-run in this system is manifested at the population level, demonstrated through *in vitro* experiments in which a smallish (circa 100 cells) cluster of NCC are cultured close to a similarly sized cluster of PCs [13]. These two clusters are subsequently observed to move in concert, maintaining a similar distance of separation as the NCC cluster persistently pursues the cluster of PCs. Given that certain pathologies may arise from failed migration of NCCs to a target tissue [14], the robustness of collective migrations of this nature may be crucial for correct tissue development.

Logically, population-level chase-and-run would seem a natural outcome of chase-and-run interactions at the individual level. However, it would seem equally plausible to suppose other possible outcomes: for example, a running cluster may escape a chasing group and move out of range, or chasers may catch the escaping group. Alternatively, individuals could be chasing one another in a range of different directions, causing the overall populations to be more-or-less homogeneously spread. The objective here is to systematically explore the various dynamics that can emerge within systems where a chase-and-run interaction occurs. To achieve this, in the next section, we will describe a somewhat minimal population-level model whereby each interaction is characterized by its type (attracting or repelling), strength and range. We show that this simple model is capable of exhibiting a broad spectrum of population-level dynamics, including those described above. We subsequently explore the conditions under which particular types of patterns can robustly emerge, focusing on the crucial role played by the interaction ranges.

2. Methods

We present our model as a system of non-local advection–diffusion PDEs for continuous population densities, a structure commonly used to phenomenologically describe interaction-based movements in cellular and animal populations (refer to [6] for a review). However, it is noted that these equations can also be derived through coarse-graining a stochastic individual (agent-based) model (refer to the electronic supplementary material); this connection is exploited later to determine the extent to which behaviours observed at the continuous (deterministic) level translate to the discrete (stochastic) level.

The model assumes that movements are governed by a set of homotypic and heterotypic interactions between neighbouring members of two populations: chasers, C , and runners, R . Neighbouring here refers to those members within a distance specific to a particular interaction. Precisely, we consider (refer to figure 1a):

- *Attracting homotypic interactions.* This could represent animal species with natural tendencies to herd or pack with conspecifics, or cell populations that attract each other, e.g. through cell–cell adhesion or ligand-mediated co-attraction.
- *Chase-and-run heterotypic interactions.* Specifically, members of C (the run) are attracted to neighbouring members of R (the chase), but members of R are repelled by neighbouring members of C .

We note that in real systems a particular interaction could be composed of both attracting and repelling components, e.g. ‘come close (attraction), but not too close (repulsion)’. Incorporating this subtlety would increase parametrization and complexity, so presently we have reduced to a net interaction: attraction or repulsion. We return to this in the discussion.

Denoting by $C(\mathbf{x}, t)$ and $R(\mathbf{x}, t)$ the two population densities at position $\mathbf{x} \in \Omega \subset \mathbb{R}^n$ and time $t \in [0, \infty)$, we consider the following system of equations:

$$\frac{\partial C}{\partial t} = \nabla \cdot \left[D_C \nabla C - \frac{C \alpha_{CC}}{V_n(\xi_{CC})} \phi(C+R) \int_{\mathcal{B}_{\xi_{CC}}^n} C(\mathbf{x} + \mathbf{y}, t) \mathbf{e}_y d\mathbf{y} - \frac{C \alpha_{CR}}{V_n(\xi_{CR})} \phi(C+R) \int_{\mathcal{B}_{\xi_{CR}}^n} R(\mathbf{x} + \mathbf{y}, t) \mathbf{e}_y d\mathbf{y} \right], \quad (2.1a)$$

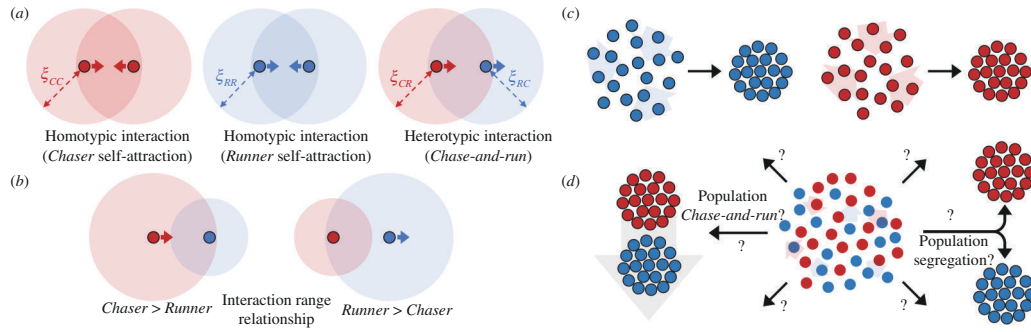


Figure 1. (a) Individual chasers (dark red circles) and runners (dark blue circles) interact across range ξ_{ij} (light coloured circles) such that (left) chasers are attracted to other chasers, (middle) runners are attracted to other runners and (right) chase-and-run describes the heterotypic interactions. (b) Distinct ranges lead to various possibilities, such as distances over which (left) chasers detect runners but not *vice versa* or (right) runners detect chasers but not *vice versa*. (c) In the absence of the other population, self-attracting interactions aggregate populations. (d) The principal question we ask in this article is as follows: in a mixed scenario, what dynamics can arise and what drives the selection of a particular dynamic? Logical reasoning would suggest possibilities that could include a segregated scenario, where the two populations separate into two essentially non-interacting groups or a population-level chase-and-run dynamic.

$$\frac{\partial R}{\partial t} = \nabla \cdot \left[D_R \nabla R - \frac{R\alpha_{RR}}{V_n(\xi_{RR})} \phi(C+R) \int_{B_{\xi_{RR}}^n} R(\mathbf{x}+\mathbf{y}, t) \mathbf{e}_y d\mathbf{y} - \frac{R\alpha_{RC}}{V_n(\xi_{RC})} \phi(C+R) \int_{B_{\xi_{RC}}^n} C(\mathbf{x}+\mathbf{y}, t) \mathbf{e}_y d\mathbf{y} \right]. \quad (2.1b)$$

The three-movement terms on the right-hand side of (2.1a) correspond to an undirected (diffusive) component, a directed movement due to homotypic interactions between members of C and a directed movement due to heterotypic interactions. Correspondingly, those on the right-hand side of (2.1b) derive from diffusive movement, homotypic interactions between members of R , and heterotypic interactions. We use B_{ξ}^n to denote the n -dimensional ball of radius ξ , $V_n(\xi)$ to denote its corresponding volume, and set $\mathbf{e}_y$ as the unit vector in the direction of $\mathbf{y}$ (if $n = 1$, this is understood to be the sign of y). Models with the above structure have been readily adopted in recent years (refer to [6] for a review), and we note in particular a number of studies for two (or more) species for both cellular [15–21] and ecological [22–26] interactions.

The direction and strength of each directed movement term is determined according to the integral evaluations of the surrounding population densities, and each requires two parameters, α_{ij} and ξ_{ij} . Parameters α_{ij} , for $ij \in \{CC, CR, RC, RR\}$, measure the strength and form of response; specifically, the response of members of population i to members of populations j . A positive (negative) value for α_{ij} indicates that i is attracted to (repelled by) j and the magnitude determines the strength of response. Accordingly, to describe our chase-and-run configuration, we concentrate on a regime in which $\alpha_{CC}, \alpha_{RR}, \alpha_{CR} \geq 0$, but $\alpha_{RC} \leq 0$. The parameters ξ_{ij} , for $ij \in \{CC, CR, RC, RR\}$, denote interaction range parameters: the maximum separation distance over which each type of interaction can occur. Therefore, a relationship $\xi_{CR} > \xi_{RC}$ implies that the range over which the chase occurs is greater than that of the run (refer to figure 1b). Note that we will often restrict to a two-dimensional parameter space (ξ_R, ξ_C), where $\xi_{RC} = \xi_{RR} \equiv \xi_R$ represents the runner interaction range and $\xi_{CR} = \xi_{CC} \equiv \xi_C$ represents the chaser interaction range. Restricting to these two parameters could be broadly interpreted as defining distinct perception ranges for each population: ξ_R (or ξ_C) represents the perception range of runners (or chasers), and subsequently defines the maximum range over which its homotypic and heterotypic interactions can form. As a note, for simplicity, the model relies on a ‘top hat’ interaction kernel (for other kernels, refer to [6] and references therein), in which the magnitude of an interaction response does not change with the separation distance (beyond individuals needing to be within the relevant range). The asymmetric interaction (the chase-and-run) coupled with the distinct interaction ranges distinguishes the present work from others noted above, which largely are restricted to symmetric interactions across identical ranges.

We note that D_C and D_R are diffusion coefficients that measure the level of random movement. The function $\phi(C+R)$ curbs directed movement if the population density becomes excessively high. This is not an issue in one dimension (1D), where we will therefore set $\phi(C+R) = 1$ for simplicity. In two dimensions (2D), however, self-attraction tends to generate highly concentrated population densities when $\phi(C+R) = 1$, and we will set $\phi(C+R) = 1/(1+C+R)$ to avoid this unwanted behaviour in 2D.

We study (2.1a) and (2.1b) in both one-dimensional and two-dimensional geometries. To minimize domain/boundary-induced effects we impose wraparound (periodic) boundary conditions, by setting the domain Ω to be a circle (of circumference L) in 1D, or torus (of dimensions $L \times L$) in 2D. Under these boundary conditions, the total population masses are conserved and uniform steady-state distributions are therefore determined by the mean initial distributions, i.e.

$$(C_s, R_s) = \left(\frac{1}{L^n} \int_{\Omega} C(\mathbf{x}, 0) d\mathbf{x}, \frac{1}{L^n} \int_{\Omega} R(\mathbf{x}, 0) d\mathbf{x} \right). \text{ Two forms of initial conditions are considered: pre-aggregated, where } C(\mathbf{x}, 0) \text{ and } R(\mathbf{x}, 0) \text{ are concentrated as Gaussian distributions; dispersed, where } C(\mathbf{x}, 0) \text{ and } R(\mathbf{x}, 0) \text{ are uniform, bar a small random perturbation. For convenience, we assume identical diffusion coefficients and mean initial densities and an } a \text{ priori scaling of length, time and densities such that } C_s = R_s = D_C = D_R = 1.$$

3. Results

3.1. Chase-and-run generates a broad spectrum of population dynamics

To motivate the investigation that follows, we pose a simple question: does chase-and-run at an individual level lead to chase-and-run at the population level? More precisely, can aggregated groups of the two populations move in synchronicity (refer to [figure 1d](#))? Logically, positive homotypic interactions should allow runners and chasers to maintain aggregated forms, and the heterotypic chase-and-run should coordinate the group movements. Simulations confirm this natural conceit, where we observe that two groups placed in proximity (such that a non-negligible level of heterotypic interactions is initially present) form a sustained synchronized movement (see [figure 2a](#)).

However, this (numerically) stable configuration is found to break down under relatively subtle variations to the interaction strengths. For other parameter combinations, we observe population segregation. Here, runners escape the interaction range of chasers and the two populations settle into fixed and separated aggregates (see [figure 2b](#)), each supported through self-attraction. An alternative form of stationary pattern occurs when chasers are able to reach and trap the runners, resulting in a single mixed aggregate (see [figure 2c](#)). More complicated dynamics arise for other parameter combinations where, rather than settling into stable chase-and-run or stationary aggregates, we observe sustained oscillations in space and time, which can be both periodic ([figure 2d](#)) or aperiodic ([figure 2e](#)). Having established that a broad spectrum of patterns are possible, we next explore how interaction range impacts on emergent dynamics in populations of chasers and runners.

3.2. Interaction ranges determine pattern selection

As noted, different interactions may operate across different ranges, e.g. if populations have distinct limits to their perception depth. We consider emergent dynamics across (ξ_R, ξ_C) parameter space from dispersed initial conditions, which allows us to determine whether a particular pattern forms without initial bias. Then, a Turing-type linear stability analysis (LSA) can be used to establish whether the uniform steady state can be unstable to inhomogeneous perturbations and patterns emerge (see electronic supplementary material). Largely speaking, homotypic attractions drive pattern formation—although we note some subtleties that emerge in 2D [20]—and allow one or both of the populations to accumulate into one or more groups. The presence of a heterotypic chase-and-run typically leads to a ‘Turing-wave’ type instability, so that emerging patterns are predicted to oscillate in both space and time; these oscillations are observable in the early time dynamics for each of the simulations plotted in [figure 3a\(ii–v\)](#).

LSA provides insight into initial pattern emergence, but for longer time dynamics we rely on numerical simulations. We numerically solve (2.1a)–(2.1b) for different (ξ_R, ξ_C) combinations and classify the dynamics at the end of each simulation. We cannot exclude that an observed pattern represents a long-time transient rather than a stable form [27], but simulations are performed for times that extend significantly beyond the characteristic timescale of pattern formation (an order of magnitude or more). [Figure 3a](#) shows the classification of parameter space, where we observe a clear demarcation into distinct regions. If both species have small interaction ranges, patterns do not form and densities evolve to the uniform steady-state solution ([figure 3a\(i\)](#))—this region coincides with the region of stability predicted by LSA. Intuitively, small interaction ranges limit the size of the observable population and restrict how much interaction-based movement can be generated. Adjacent to this region we observe sustained spatiotemporal oscillations. These transform from periodic and low amplitude ([figure 3a\(ii\)](#)) to aperiodic and large amplitude ([figure 3a\(iii\)](#)) with increasing distance from the stability/instability boundary; this coincides with increased pattern growth rates and an increase in the number of unstable modes, predicted by the LSA (see electronic supplementary material).

Setting even larger interaction ranges results in patterns that initially oscillate before transitioning into one of two general forms: a population-level chase-and-run ([figure 3a\(iv\)](#)) or stationary aggregates ([figure 3a\(v\)](#)). Significantly, the regions where these two pattern types develop depend on the interaction range relationship. Broadly speaking, population chase-and-run emerges when the chasers have a (sufficiently) larger interaction range than the runners, whereas stationary aggregates form when runners have the larger interaction range. This division is more strongly accentuated when higher interaction strengths are used (see electronic supplementary material, [figure S3a](#)), where we find chase-and-run ($\xi_C > \xi_R$) or stationary aggregates ($\xi_C < \xi_R$) and greatly reduced regions of oscillating or uniform solutions. To test whether it is the heterotypic interactions that drive this pattern selection, we see a similar division when only (ξ_{CR}, ξ_{RC}) are varied: $\xi_{CR} > \xi_{RC}$ ($\xi_{RC} > \xi_{CR}$) biases pattern selection to a chase-and-run (stationary) form (electronic supplementary material, [figure S3b](#)).

To probe further how the nature of the chase-and-run interactions drive the dynamics, we investigate pattern selection in each of two regimes $\xi_C > \xi_R$ and $\xi_R > \xi_C$, while changing $(\alpha_{RC}, \alpha_{CR})$. This allows a test into the robustness of an observed dynamic while modulating the strength of the heterotypic interactions. Corroborating the key importance of interaction range, we see a robust emergence of chase-and-run across a wide range of interaction strengths, if $\xi_C > \xi_R$ ([figure 3b](#), left panel). Similarly, we see robust emergence of stationary aggregates when $\xi_R > \xi_C$ ([figure 3b](#), right panel). These explorations also highlight some further subtleties. For example, stationary aggregates are found to range from completely segregated (clusters of only chasers or only runners), to mixed/segregated (some clusters contain both chasers and runners), to completely mixed (the two populations are co-localized) (see electronic supplementary material, [figure S3c](#)). This transition follows changing strengths of the heterotypic interactions: $\alpha_{RC} > \alpha_{CR}$ can allow runners to completely escape chasers and form a separate group, while $\alpha_{CR} > \alpha_{RC}$ can allow chasers to catch and trap runners. Modulations to the chase-and-run can also occur. For example, lower run interaction strength

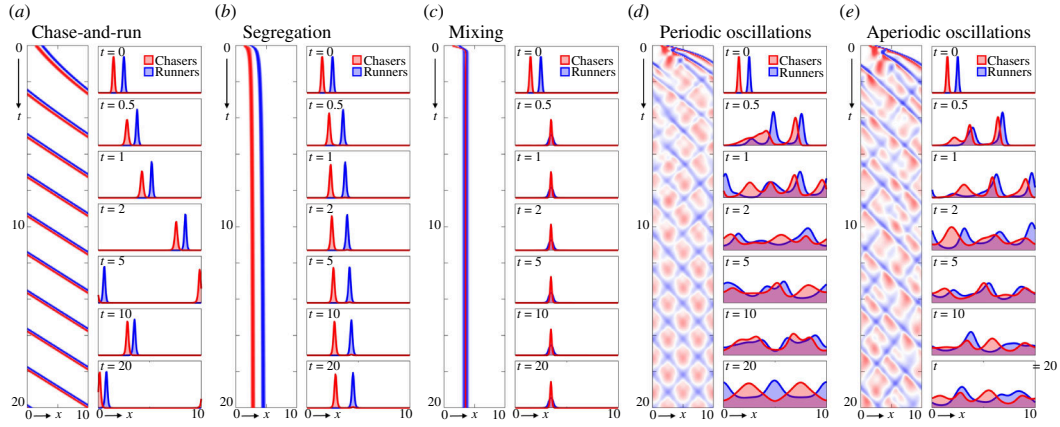


Figure 2. Dynamics of the chase-and-run system (2.1a)–(2.1b) in 1D on $[0, 10]$: each panel shows chasers (red) and runners (blue) on a kymograph and distributions at the stated times. Here, interaction ranges $\xi_{RR} = \xi_{CR} = \xi_{RC} = \xi_{RR} = 1$ and interactions strengths $\alpha_{CC} = \alpha_{RR} = 3$ with: (a) $(\alpha_{CR}, \alpha_{RC}) = (1, -1)$, (b) $(\alpha_{CR}, \alpha_{RC}) = (2, -3)$, (c) $(\alpha_{CR}, \alpha_{RC}) = (2, -1)$, (d) $(\alpha_{CR}, \alpha_{RC}) = (5, -5)$ and (e) $(\alpha_{CR}, \alpha_{RC}) = (4, -4)$. Note that $D_C = D_R = 1$ with initial conditions $C(x, 0) = C^* \exp(-50(x - 1.5)^2)$, $R(x, 0) = R^* \exp(-50(x - 2.5)^2)$, where C^* and R^* are set such that $C_s = R_s = 1$. For details of the numerical scheme, refer to the electronic supplementary material.

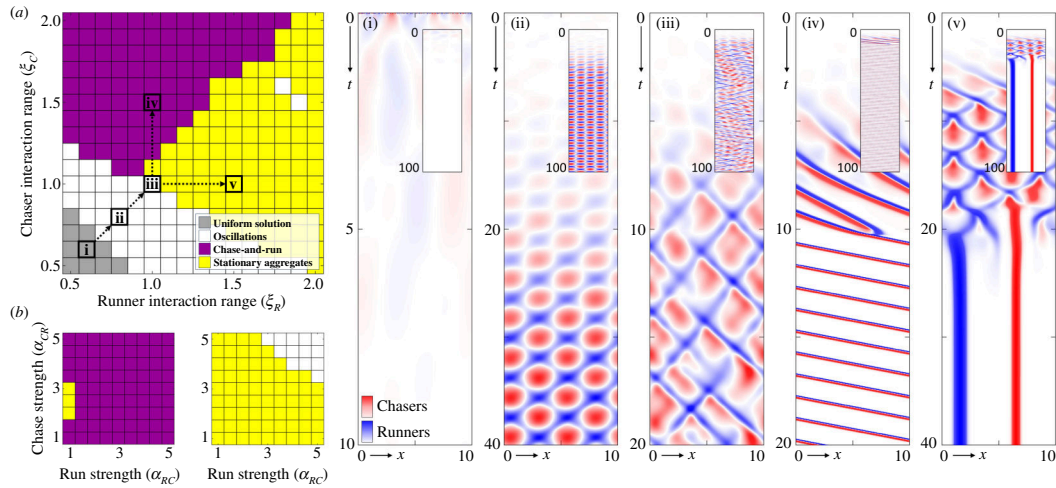


Figure 3. (a) Pattern selection across interaction range space. We classify the pattern for each (ξ_R, ξ_C) pair as: uniform solution (grey), spatiotemporal oscillations (white), population chase-and-run (magenta) and stationary aggregates (yellow). The kymographs plotted in (i–v) illustrate representative solutions for certain (ξ_R, ξ_C) pairs, as indicated in (a). For other model parameters, we fix $\alpha_{CC} = \alpha_{CR} = -\alpha_{RC} = \alpha_{RR} = 3$, $D_C = D_R = 1$, $L = 10$ and initial conditions in the dispersed form, such that $C_s = R_s = 1$. (b) Pattern selection across interaction strength space. Here, the pair $(\alpha_{RC}, \alpha_{CR})$ is varied, while interaction ranges are selected (left) from the chase-and-run region (specifically, $(\xi_R, \xi_C) = (1, 1.5)$) and (right) from the stationary aggregates region (specifically, $(\xi_R, \xi_C) = (1.5, 1)$). We fix $D_C = D_R = 1$, $L = 10$ and set initial conditions in the dispersed form. Specifically, we set $C(x, 0) = 1 + r_1(x)$ and $R(x, 0) = 1 + r_2(x)$, where $r_1(x), r_2(x) \in [-0.1, 0.1]$ denote noises added to the uniform steady-state densities $(C_s, R_s) = (1, 1)$.

results in some runners being left behind; on the slightly artificial periodic domain, these runners are subsequently recollected and the process repeats (electronic supplementary material, figure S3d).

Summarizing, we find that distinct interaction ranges have a strong influence on emergent population-level dynamics under chase-and-run type interactions.

3.3. Chase-and-run in the plane

In 2D a runner can escape a chaser across all angles of the circle. We therefore investigate how dynamics change for populations distributed across $\Omega = [0, L] \times [0, L]$, under the same general homotypic and heterotypic interactions. Populations are initially dispersed (figure 4b) and we again explore (ξ_R, ξ_C) -space. Simulations in this section are accompanied by electronic supplementary material animations.

Principally, we observe the same separation of parameter space into sustained population-level chase-and-run (when $\xi_C > \xi_R$) or stationary aggregates (when $\xi_R > \xi_C$) (see figure 4a). Consistent with 1D, a more defined separation emerges with higher interaction strengths: lower interaction strengths lead to smaller regions of chase-and-run or stationary aggregates (see electronic supplementary material, figure S4a, top), and larger regions of oscillating (refer to electronic supplementary material, figure S4b(i) for an example) or uniform solutions. While chase-and-run is implemented through the heterotypic interactions,

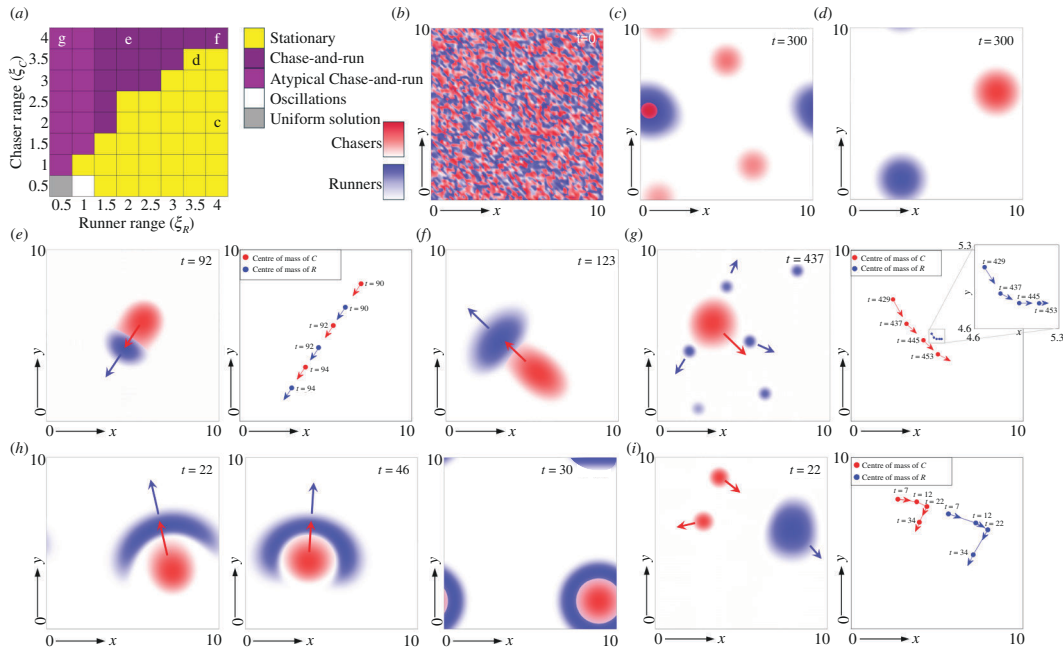


Figure 4. (a) Classification of (ξ_R, ξ_C) parameter space, for simulations of (2.1a)–(2.1b) in 2D. (b) Dispersed initial conditions at $t = 0$, specifically $C(x, y, 0) = 1 + r_1(x, y)$ and $R(x, y, 0) = 1 + r_2(x, y)$, where $r_1(x, y), r_2(x, y) \in [-0.1, 0.1]$ denote noises added to the uniform steady state $(C_s, R_s) = (1, 1)$. (c) Stationary aggregate solution found at point c in (a). (d) Stationary aggregate solution found at point d in (a). (e, left) Stereotypical chase-and-run found at point e in (a) (see electronic supplementary material, movie S1); (e, right panel) the positions of the centre of masses for chasing and running groups over a range of times. (f) Chase-and-run, with interaction ranges found at point f in (a) (see electronic supplementary material, movie S2). (g, left panel) Atypical chase-and-run found at point g in (a) (see electronic supplementary material, movie S3); and (g, right panel) centre of masses for the chasing and the running groups. (h) Crescent-shaped runners for $\xi_{CC} = \xi_{CR} = \xi_{RR} = 4$ and (left) $\xi_{RC} = 2$, (middle) $\xi_{RC} = 1$ and (right) $\xi_{RC} = 0.5$ (see electronic supplementary material, movies S4–S6). (i, left panel) Atypical chase-and-run for $\xi_{CR} = \xi_{RC} = \xi_{RR} = 4$ and $\xi_{CC} = 1$ (see electronic supplementary material, movie S7); and (i, right panel) centre of masses for the chasing and the running groups. All simulations in this figure use $\alpha_{CC} = \alpha_{RR} = \alpha_{CR} = -\alpha_{RC} = 15$, $D_C = D_R = 1$, $L_x = L_y = 10$ and $\phi(C + R) = 1/(1 + C + R)$.

increasing the chase-and-run strength alone does not encourage the emergence of these dynamics at the population level; in contrast, we see an expansion of the oscillating regime (refer to electronic supplementary material, figure S4a, bottom). This reinforces that both homotypic and heterotypic interactions play a critical role in setting the population-level dynamics.

Consistent with 1D, we find that stationary aggregates range from mixed (figure 4c) to segregated (figure 4d). In regions of chase-and-run, homotypic self-attraction first organizes the dispersed populations into clusters. Then, heterotypic interactions drive the cluster of chasers to pursue the cluster of runners, maintaining consistent shape and speed (figure 4e, electronic supplementary material, movie S1). This stereotypical chase-and-run robustly emerges when self-interaction strengths are sufficiently strong, and on larger domains (electronic supplementary material, figure S4c). However, some subtler features become apparent according to the runner interaction range. Under more-or-less equal interaction ranges the two groups are roughly equal in size (figure 4f, electronic supplementary material, movie S2). Reducing the runner interaction range groups runners into a smaller (and more concentrated) group (figure 4e). Below some critical value, however, runners instead organize into multiple small groups. Run-and-chase still occurs, but in an atypical form: chasers continue to pursue runners, but with multiple runner clusters the chasing direction changes over time (figure 4g, electronic supplementary material, movie S3). With its limited interaction range, a cluster of runners remains fixed in position until a chasing group is sufficiently close, at which point it moves away.

To investigate further patterning subtleties, we perturbed specific interaction range parameters and commented on notable instances. Lowering only the run interaction range (ξ_{RC}) leads to a crescent-shaped configuration ahead of the chasing group (figure 4h, electronic supplementary material, movies S4 and S5). Here, runners only detect the nearest portion of the chasing group and, consequently, the escape direction can vary significantly across the runner population. Self-attraction of runners (ξ_{RR}) occurs over longer range, maintaining cohesion and generating the crescent. When ξ_{RC} decreases below a critical value, runners entirely surround chasers in an annulus-like shape, so that net motion is prevented (figure 4h, electronic supplementary material, movie S6). Finally, reducing the chaser self-attraction range (ξ_{CC}) below a critical value results in multiple (smaller) chaser groups. At a phenomenological level, the subsequent dynamics can be viewed as a reversal of the atypical chase-and-run above, where now a single group of runners is chased by multiple groups of chasers (figure 4i, electronic supplementary material, movie S7). These results highlight that the individual homotypic interaction ranges strongly influence the spatial extension of groups at the population level.

Broadly speaking, investigations in 2D reinforce that interaction ranges dictate the emerging dynamics. Yet, we also observe increasingly nuanced patterning, suggesting that greater control is required if a particular pattern is to emerge in higher dimensions.

3.4. Individual-based modelling

The continuous model is predicated on an assumption that the populations can be approximated by density distributions, rather than individual positions. The extent to which this is valid for the population sizes encountered in real-world instances of collective movement is uncertain, and we now determine whether behaviours observed for equations (2.1a) and (2.1b) carry to an individual-based model (IBM). The lattice-based IBM is described in the electronic supplementary material, and features two populations of particles which interact according to the homotypic and heterotypic interactions described previously. Notably, coarse-graining the IBM leads to the continuous formulation given by equations (2.1a) and (2.1b) equations 2.1a, 2.1b (refer to the electronic supplementary material). Consequently, an IBM parameter set (including lattice spacing and time step) can be chosen in correspondence to a given simulation of the continuous model. Note that for all IBM simulations, we use 100 individuals per populations (consistent with typical sizes of populations of NCCs and PCs in the example cited in the introduction).

For the 1D IBM (refer to the electronic supplementary material), we consider a lattice of length $L = 10$ with spacing $l = 0.1$. Setting the time step $\tau = 0.005$ then ensures $D = 1$, as used in simulations of the continuous models. As before, we keep $\alpha_{RC} < 0$ and $\alpha_{CC}, \alpha_{CR}, \alpha_{RR} > 0$, so that individuals of population C chase those of population R. In figure 5a, we classify the type of pattern observed following simulations of the IBM, covering the same parameter space as in figure 3a. Consistent with the continuous model, we see sustained chase-and-run when the interaction range of chasers is large compared to those of the runners (figure 5b), otherwise, we see more-or-less stationary aggregates (figure 5c); there can be some ‘wobble’ in a stationary aggregate position over time, due to inherent stochasticity, but there is no clear chasing. Finer points of detail also translate, such as the range of stationary aggregate patterns from completely segregated to mixed. We further confirmed through simulations of a 2D IBM, using equivalent settings to simulations of the continuous models reported in figure 4. A population chase-and-run behaviour can be observed when $\xi_C > \xi_R$ (figure 5d) and stationary aggregates for $\xi_R > \xi_C$ (figure 5e).

However, some noteworthy differences emerge. First, for the IBM we have not observed (clear) patterns representing homogeneity or sustained oscillations, at least not when using parameter settings equivalent to those within the continuous simulations. This perhaps suggests that these patterns are more reliant on the assumption that a population can be approximated by a continuous distribution, rather than individual positions. Also, in 2D simulations we have found chase-and-runs where the direction of group movements changes over time: at lower population sizes, the occasional loss of individuals from runner or chaser groups can prove influential, creating ‘rogue’ individuals that can subsequently alter cluster dynamics.

3.5. Self-attraction is required for robust chase-and-run

We have observed how population chase-and-run emerges in 1D, 2D and in IBM simulations, when the range of the chase is greater than that of the run. However, all simulations to date assume that each population has a self-attraction that allows their self-aggregation. We investigate whether this is crucial for the emergence and maintenance of a pattern type.

First, we repeat the 1D analysis, investigating pattern selection in (ξ_R, ξ_C) -space under two cases: (i) only runners self-aggregate ($\alpha_{RR} > 0$, $\alpha_{CC} = 0$); and (ii) only chasers self-aggregate ($\alpha_{CC} > 0$, $\alpha_{RR} = 0$). In both instances, we (broadly) find the same behaviour: across a large region of parameter space corresponding to $\xi_C > \xi_R$, a sustained chase-and-run emerges at the population scale (figure 6a,b). Self-aggregation in the runners leads to their clustering which—through the heterotypic interactions—collect chasers into a looser group in the rear (figure 6c, top). Self-aggregation in the chasers leads to a chaser cluster that ‘herds’ runners ahead (figure 6c, bottom).

However, these observations of emergent and sustained chase-and-run in 1D evaporate in higher dimensions. When only runners aggregate, scenarios of chase-and-run ($\xi_C > \xi_R$) become ‘incoherent’, with the groups repeatedly splitting and reforming over time (figure 6d, refer to electronic supplementary material, movie S8). Here, the somewhat dispersed chasing group transmits variable directional information into the cluster of runners, inducing their escape into directions that can split the group; the reformation of groups can be part-attributed to the peculiarities of the boundary conditions. When only chasers aggregate, a $\xi_C > \xi_R$ scenario leads to a stationary cluster of chasers surrounded by uniformly dispersed runners (figure 6e). Unlike in 1D, where runners can be herded and trapped by the geometry, different escape directions mean that runners spread until chasers have no coherent direction of pursuit.

To test this under a more controlled setting, chasers and runners are initialized as juxtaposed clusters, figure 6f: this could be regarded as an optimal initial state to encourage chase-and-run. Also, to limit the influence of boundary conditions we consider a larger domain. Setting $\xi_C > \xi_R$, if both populations self-aggregate we observe robust and coherent chase-and-run in (seeming) perpetuity (figure 6f(i)). When only runners self-aggregate a transient chase-and-run can occur, but the runner and chaser groups lose mass over time and eventually chase-and-run disintegrates (figure 6f(ii), refer to electronic supplementary material, movie S9). When only chasers self-aggregate, any chase-and-run is extremely brief: the runners quickly disperse in all directions to leave a stranded group of chasers (figure 6f(iii), see electronic supplementary material, movie S10). Similarly, when neither population can self-aggregate any coherent motion is transient and the populations disperse to the uniform steady state (figure 6f(iv)). Finally, we investigate dynamics for populations that are initially arranged into stripes (figure 6g), for the same interactions as in figure 6f(i). At first, the populations move in stripes of chasers and runners, before the front destabilizes and resolves into discrete clusters that undergo separate chase-and-run. This clustering principally stems from the self-attraction terms that round populations into compact groups.

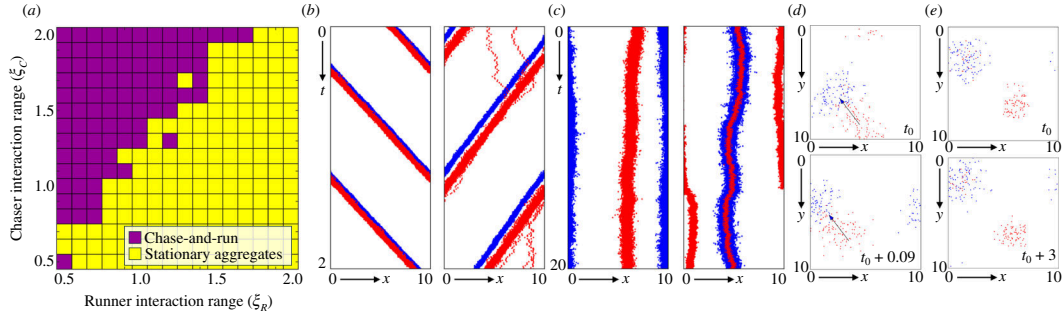


Figure 5. Simulation output of the IBM. (a–c) 1D simulations. (a) Classification of (ξ_R, ξ_C) space analogous to figure 3a. (b) Representative examples of chase-and-run dynamics in the IBM. (c) Representative examples of stationary aggregate dynamics in the IBM. (d,e) 2D simulations. (d) 2D chase-and-run, analogous to figure 4e with plots showing chasers (red) and runners (blue) at a reference time (top) t_0 and (bottom) $t_0 + 0.09$. (e) 2D stationary aggregates analogous to figure 4c with plots at a reference time (top) t_0 and (bottom) $t_0 + 3$.

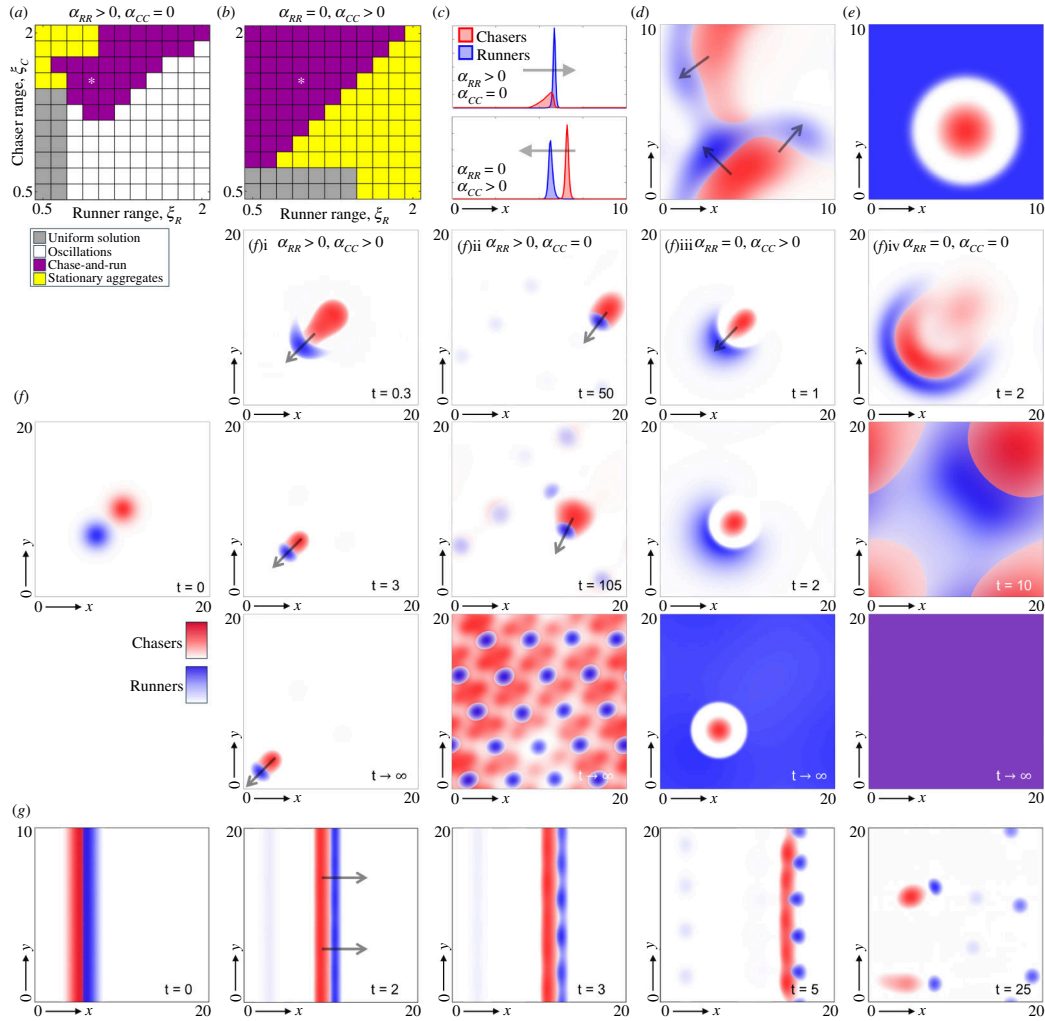


Figure 6. Dynamics for various self-aggregating tendencies. (a,b) Classification of (ξ_R, ξ_C) -space for equations (2.1a) and (2.1b) in 1D with $\alpha_{CR} = 3, \alpha_{RC} = -3$ and $(\alpha_{RR}, \alpha_{CC}) =$ (a) (3,0), (b) (0,3). (c) Typical chase-and-run profiles in 1D, for positions marked with the white star in (a,b). (d) Irregular chase-and-run in 2D when only runners self-aggregate, from dispersed initial conditions (electronic supplementary material, movie S8). Here, $\xi_C = 4, \xi_R = 3, \alpha_{RR} = \alpha_{CR} = 15, \alpha_{RC} = -15, \alpha_{CC} = 0$. (e) Stationary aggregate of chasers in 2D when only chasers self-aggregate. Here, $\xi_C = 4, \xi_R = 2, \alpha_{CC} = \alpha_{CR} = 15, \alpha_{RC} = -15, \alpha_{RR} = 0$. (f) Pre-aggregated initial conditions $C(x, y, 0) = C^* \exp(-0.2(x - 10)^2 - 0.2(y - 10)^2)$, $R(x, y, 0) = R^* \exp(-0.2(x - 7)^2 - 0.2(y - 7)^2)$, where C^* and R^* are such that $C_s = R_s = 1$ (left panel). Columns from left to right show snapshots for $(\alpha_{CC}, \alpha_{RR}) =$ (i) (15, 15), (ii) (0, 15) (electronic supplementary material, movie S9), (iii) (15, 0) (electronic supplementary material, movie S10), (iv) (0, 0). Other parameters set at $\xi_C = 4, \xi_R = 2, \alpha_{CR} = 15, \alpha_{RC} = -15$ and ' $t \rightarrow \infty$ ' indicates the long-time pattern form. (g) Populations initially arranged into parallel stripes, specifically a small random perturbation of $C(x, y, 0) = C^* \exp(-(x - 5)^2)$, $R(x, y, 0) = R^* \exp(-(x - 6)^2)$, where C^* and R^* are such that $C_s = R_s = 1$. Other parameter values are $\alpha_{CC} = \alpha_{CR} = -\alpha_{RC} = \alpha_{RR} = 15, \xi_C = 4, \xi_R = 2$. All simulations use $D_C = D_R = C_s = R_s = 1$.

4. Discussion

We have explored the collective dynamics that emerge within mixed populations of chasers and runners, where the two populations have an inherent tendency to aggregate with members of their own kind and pursue and evade the other population, respectively. Focusing on the interaction ranges, we observe two dominating pattern forms: (i) when runners interact over larger distances than chasers (or the range of the run is greater than that of the chase), quasi-stationary and separated clusters emerge that undergo negligible interaction; (ii) when chasers interact over larger distances than runners (or the range of the chase is greater than that of the run), a collective chase-and-run forms where the cluster of chasers pursues the cluster of runners. Given sufficiently strong interaction strengths, this overall observation is apparently robust with respect to moderate parameter variations, when extended into a higher dimension setting and when examined within the finite population sizes and stochastic setting of a corresponding IBM. As such, distinct interaction ranges between species appear to significantly determine emergent collective dynamics.

Intentionally, we have formulated the model in a minimal and generic manner, so that (in principal) the results could be interpreted for cellular or animal populations. For the former, chase-and-run dynamics have been observed in zebrafish pigment cells [12] and *Xenopus laevis* NCC–PC populations [13]. The latter forms a particularly apposite example, as *in vitro* studies reveal an analogous phenomenon of population-level chase-and-run [13]: NCCs form the chasing group; and PCs the running group. Experiments have uncovered various cellular interactions, including contact inhibition of locomotion (CIL) [13,28] between all cells: the touching of two cells reverses their movement directions. Consequently, CIL forms a repulsion between all cells, but other interactions may switch some of these to attracting. First, PCs adhere to each other (through E-cadherin), which overcomes CIL to allow PC-to-PC attraction [13]. Second, attracting homotypic interactions also occur between NCCs [29,30], seemingly autotaxis-mediated through a secreted attractant. Third, NCCs are attracted to PCs through the latter secreting a chemotaxis factor (Sdf-1) [13]. Under the reductive reasoning of a considered ‘net interaction’, it is plausible that the set of interactions may conform to those explored here and we predict that a robust population chase-and-run will arise if NCCs interact over larger ranges. Indeed, this is quite plausible: PC interactions are seemingly through direct contact (membrane–membrane binding), while NCC interactions involve diffusing ligands. The simulations presented in figure 6f are particularly apposite for this application, as the *in vitro* experiments place NCCs and PCs as juxtaposed clusters (akin to the $t = 0$ setting of those simulations). While indefinite chase-and-run requires both populations to have self-attraction, long transient chase-and-runs can still occur when only the runners have (strong) self-attraction: if this transient is much longer than the timescale of experiments, this may be sufficient. Thus, our simulations would suggest that while strong self-attraction is essential in the runner group (i.e. the PCs), a similar requirement may not be strictly necessary for the chasers (i.e. the NCs); possibly this could be tested through knocking out the mechanism of co-attraction between NCCs.

Initializing as clusters recapitulates an *in vitro* setting. The neural crest itself forms along the dorsal neural tube, creating a ‘stripe’ along the vertebrate embryo. Experimental observations of the *Xenopus* NCC–PC system—and simulations of an ABM—have revealed that as the NCCs move away from the crest, they insert into alternating ‘streams’ with the surrounding PCs [31]. In other words, a form of symmetry-breaking takes place along the length of the (cranial) neural crest. Our simulations with a stripe setting (figure 6g) also show a symmetry-breaking, with the stripe destabilizing into multiple clusters. We do not, however, observe a formation of alternating streams. This is not in itself surprising, given that our (current) model is not explicitly tailored to describe this particular application. A more considered investigation would be a very interesting avenue, necessitating extension to a ‘next level’ model, for example: dropping the reduction to a net interaction, so that the relationship between two cell types may be composed of both repulsive (e.g. CIL) and attractive (e.g. co-attraction) interactions; explicitly modelling key extracellular factors (e.g. Sdf-1), receptor–ligand interactions, etc.; parametrizing the model as far as possible according to data. This would offer a complementary approach for modelling the *Xenopus* NCC–PC system, as previous approaches have been restricted to agent-based models [31,32].

In ecology, our chase-and-run can be viewed analogously to a pursuit and escape interaction and the model described here could help in understanding the spatial arrangements of predators and prey on a landscape. The concept of a ‘landscape of fear’, whereby prey avoid areas they believe predators to be living, has gained much attention in recent years [33,34]. Improvements in animal tracking technology [35] have begun to allow researchers to map such landscapes and rigorously demonstrate prey avoidance [36,37]. Likewise, the notion of prey-taxis [38], whereby predators seek to locate themselves near prey, is well-understood from both empirical [39] and mathematical [40] perspectives. Both aspects have been studied mathematically in a model similar to ours but with quadratic rather than linear diffusion [41]. However, in empirical situations, it is less clear whether a combination of predator-avoidance and prey-taxis behaviour ever actually leads to the kind of population-level chase-and-run described here. Indeed, empirical evidence suggests that most animals tend to locate themselves in relatively stationary home ranges rather than moving over the landscape in response to predation pressure or seeking out mobile prey [42]. A possible reason for this is that landscape heterogeneity might ‘pin’ the population-level movements, so that, for example, prey might spend time in a woodland area where they can hide from predators, whereas predators might live close to the forest in hope of catching any prey that venture out [43]. Mathematical investigation of individual-level chase-and-run within heterogeneous environments would help understand why we still tend to see predators and prey using relatively stationary home ranges, despite the theoretical possibility of population-level chase-and-run in homogeneous environments.

Notions of followers and leaders have gathered considerable traction in recent years [3,44–47]. The terms ‘followers’ and ‘leaders’ imply a hierarchy—the latter guides the former—though in some cases their designations may simply reflect a spatial position [48]: leaders at the front, followers at the rear. Placed in the context of the collective run-and-chase that emerges here, the macroscopic perspective is runners at the front and chasers at the rear. However, regarding runners as leaders and chasers as followers is clearly misleading: while the groups sort into a structured arrangement, there is no overall guidance and the net

movement (from unbiased initial conditions) will be at random. This susceptibility is highlighted by the (2D) IBM simulations, where the chase-and-run direction was deflected by ‘rogue’ individuals and/or stochastic fluctuations. The addition of a further directional bias—such as a chemoattractant—could confer a robustness to the global direction of movement and, in this regard, it may be illuminating to observe how robustness alters according to which population detects the global cue: under what circumstances do runners or chasers make the more effective leaders?

Coherent chase-and-run—by which we mean a compact cluster of chasers pursuing a compact cluster of runners—was found to emerge in both 1D and 2D. In 1D, this is particularly robust, given the stipulated relationship between interaction ranges, within both continuous and IBM formulations. Of course, this robustness can be partly attributed to the peculiarities of a 1D geometry, where runners can only escape in one of two possible directions. Away from singular or engineered instances—such as cells restricted to lines along microfabricated surfaces [49]—movements take place in higher dimensions, e.g. 2D for cells plated on a surface or animal movements on the plane. Despite the greater freedom movement direction, coherent chase-and-run does emerge robustly in 2D, but with caveats: for example (in continuous simulations), ‘complex’ chase-and-runs where the dynamics of a single large cluster of runners (chasers) is driven by multiple smaller groups of chasers (runners), or (in IBM simulations) with occasional fluctuations in the movement direction. Emergence of a particular dynamic in 2D may therefore require additional ‘control’, e.g. more tightly controlled parameters or reinforcement through environmental heterogeneity. A number of modelling and experimental studies have suggested that microenvironmental factors may indeed play an important role in confining collective migration of NCCs along particular pathways [50,51].

We have observed even more diverse pattern forms in certain regimes, including a menagerie of oscillatory dynamics. These dynamics were robust—in the sense that they continued indefinitely and persisted under perturbation—and therefore seemingly represent a stable pattern outcome of the continuous model, rather than a transient dynamic. A deeper analytical study into the patterns that arise as parameters are varied could be achieved through nonlinear bifurcation analyses [52], a possible future direction. However, it is noted that in the IBM these oscillatory dynamics were conspicuous by their absence: under equivalent parameters, solutions evolved to either the population chase-and-run or stationary aggregate patterns. The continuous model assumes that populations can be approximated by a continuous density rather than individual particle positions, and it is possible that simulations of the IBM under larger populations would reveal similar phenomena. It should also be noted that an IBM will introduce other forms of artificiality—for example, the IBM here is based on point particles that make instantaneous jumps in individual location—which could generate particularly high levels of stochasticity. Convergence to the continuous model involves various limits (e.g. smaller steps, higher population sizes), and a more detailed investigation into the correspondence between the IBM and continuous model may pinpoint scenarios under which particular outcomes may be expected to arise. It may also be worth investigating whether different IBM frameworks—for example, those where particles move continuously through space—can be scaled to the same continuous model, and whether those lead to different behaviours. Overall, though, the discrepancies we have observed reinforce a general lesson that caution should be adopted regarding ‘exotic’ patterning phenomena observed in models, particularly when found in restricted regions of parameter space.

A number of earlier studies have used similar modelling frameworks to understand the distinct patterns that can arise from self and cross interactions within heterogeneous systems. Several of these have been motivated by adhesive cell sorting, for example, [15,17–21,53]. Here, homotypic and heterotypic interactions are typically non-negative and studies have focused on how distinct adhesions lead to distinct arrangements, consistent with classical predictions of the differential adhesion hypothesis [54]. A generalization to interactions that range from attracting to repelling was conducted by Painter *et al.* [16], although the subsequent analysis was primarily restricted to simple 1D geometries and equivalent interaction ranges. Recently, Jewell *et al.* [20] have extended this to 2D, where, surprisingly, a pure chase-and-run interaction (with no homotypic interactions) was shown to be capable of generating pattern formation in 2D (and higher dimensions). Significantly, unequal interaction ranges form a requirement and the study here further reinforces the critical role that interaction ranges can play in patterning dynamics.

We have shown that wide-ranging collective dynamics can emerge, even for just two species undergoing chase-and-run interactions. Yet, the relatively simple assumption of distinct interaction ranges provides a powerful point of control, strongly determining pattern selection. Clearer understanding into the ranges over which particular interactions are mediated would provide valuable information into the organization of complex systems.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. Code is available at [55].

Supplementary material is available online [56].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. K.J.P.: conceptualization, formal analysis, investigation, methodology, writing—original draft, writing—review and editing; V.G.: formal analysis, investigation, methodology, writing—review and editing; J.R.P.: formal analysis, investigation, methodology, writing—review and editing; S.B.: formal analysis, investigation, methodology, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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